Enhanced Tomographic Assessment to Detect Corneal Ectasia Based on Artificial Intelligence

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• PURPOSE: To improve the detection of corneal ectasia susceptibility using tomographic data.

• DESIGN: Multicenter case-control study.

• METHODS: Data from patients from 5 different clinics from South America, the United States, and Europe were evaluated. Artificial intelligence (AI) models were generated using Pentacam HR (Oculus, Wetzlar, Germany) parameters to discriminate the preoperative data of 3 groups: stable laser-assisted in situ keratomileusis (LASIK) cases (2980 patients with minimum follow-up of 7 years), ectasia susceptibility (71 eyes of 45 patients that developed post-LASIK ectasia [PLE]), and clinical keratoconus (KC; 182 patients). Model accuracy was independently tested in a different set of stable LASIK cases (298 patients with minimum follow-up of 4 years) and in 188 unoperated patients with very asymmetric ectasia (VAE); these patients presented normal topography (VAE-NT) in 1 eye and clinically diagnosed ectasia in the other (VAE-E). Accuracy was evaluated with ROC curves.

• RESULTS: The random forest (RF) provided highest accuracy among AI models in this sample with 100% sensitivity for clinical ectasia (KC+VAE-E; cutoff 0.52), being named Pentacam Random Forest Index (PRFI). Considering all cases, the PRFI had an area under the curve (AUC) of 0.992 (94.2% sensitivity, 98.8% specificity; cutoff 0.216), being statistically higher than the Belin/Ambrósio deviation (BAD-D; AUC = 0.960, 87.3% sensitivity, 97.5% specificity; P = .006, DeLong's test). The optimized cutoff of 0.125 provided sensitivity of 85.2% for VAE-NT and 80% for PLE, with 96.6% specificity.

• CONCLUSION: The PRFI enhances ectasia diagnosis. Further integrations with corneal biomechanical parameters and with the corneal impact from laser vision correction are needed for assessing ectasia risk. (Am J Ophthalmol 2018;195:223–232. © 2018 Elsevier Inc. All rights reserved.)

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THE DIAGNOSIS OF ECTASIA, ESPECIALLY IN ITS milder forms, is extremely important when screening for laser vision correction (LVC), because patients with this condition are at very high risk for developing progressive ectasia and worsening of vision after the procedure.[1,2] In fact, LVC is among the most performed surgical procedures worldwide. It provides remarkable benefits in patients' quality of life with relatively low risks. Nevertheless, considering the elective nature of the procedure, potential complications must be carefully considered, especially iatrogenic ectasia.[3,4]

Iatrogenic ectasia occurs because of a biomechanical failure of corneal tissue in which its shape can no longer be maintained. This is basically related to the preoperative predisposition of the cornea and/or the impact from the procedure. The first report of progressive ectasia after laser-assisted in situ keratomileusis (LASIK) refers to a case with subclinical topographic (referred to as fruste) keratoconus.[2] In such previously susceptible (biomechanically weak) cases, even a procedure causing relatively low impact to the cornea could induce ectasia progression. Conversely, the removal of corneal tissue by LVC can induce substantial biomechanical weakening that by itself decompensates even a previously normal cornea.[5,6] In

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addition, a subsequent biomechanical stress, such as vigorous eye rubbing, could also result in chronic biomechanical failure.[7,8]

The main challenge in preventing iatrogenic ectasia is to detect cases with highest risk for developing ectasia despite a relatively low-impact procedure and no extra biomechanical stress postoperatively.[3,4] While corneal topography has unquestionably improved the sensitivity to detect abnormalities in cases with normal slit-lamp biomicroscopy and normal corrected distance visual acuity (CDVA),[9] some cases are still hard to identify in standard preoperative screening because of little to no alteration in the anterior corneal surface and/or corneal thickness.[10]

Corneal tomography with 3-dimensional geometric characterization promotes an additional understanding of corneal structure.[11] However, the complexity and amount of available parameters make proper interpretation difficult.[12]

In this scenario, machine-learning techniques with pattern recognition are very useful. Computer algorithms (or artificial intelligence [AI]) can help our cognitive ability to deal with massive amounts of information and use it appropriately. This is a constantly growing field in medical diagnosis and treatment. Its first applications in ectasia diagnosis date from the mid-1990s.[13,14] In this study we have tested AI techniques to determine their ability to detect ectasia risk by differentiating susceptible cases, early or subclinical disease, clinical keratoconus, and stable eyes.

METHODS

THIS MULTICENTER RETROSPECTIVE CASE-CONTROL STUDY followed the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board and Human Ethics Committee from the Federal University of São Paulo (UNIFESP, SP, Brazil). All patients provided informed consent for the use of their de-identified data in scientific research.

• **PARTICIPANTS:** A total of 3693 patients were enrolled from Optical Express, Glasgow, United Kingdom (clinic 1), Instituto de Olhos Renato Ambrósio, Rio de Janeiro, Brazil (clinic 2), Vincieye Clinic, Milan, Italy (clinic 3), Price Vision Group, Indianapolis, Indiana, USA, (clinic 4), and London Vision Clinic, London, United Kingdom (clinic 5). Owing to the complex nature of ectasia as a complication of refractive surgery, the preoperative data of patients with post-LASIK ectasia (PLE) were collected individually from several surgeons around the world and their identities were requested to be kept private. Surgeries in this group were performed between 2006 and 2010. The characteristics of the surgical procedure from all clinics are summarized in Table 1. Patients were divided into 2 data sets. One was used to train the model with internal cross-validations and the other was used to independently test the generated models.

The training set was composed of 3 groups. The stable group, or controls, included the preoperative status from 1 eye randomly selected from 2980 patients with stability documented for at least 7 years (Stable Train) from clinic 1 (surgeries performed between 2008 and 2009). The ectasia cases included 1 eye randomly selected from 182 bilateral clinical keratoconus (KC) patients from clinics 2 and 3, and from the preoperative data of 71 eyes from 45 patients that developed PLE.

The test set was composed of cases not included in the training model. The control test group was composed of the preoperative data of 298 patients with stable corneas for at least 4 years (Stable Test) from clinics 4 and 5 (surgeries performed between 2009 and 2012). The ectasia test group included both eyes from 188 patients with very asymmetric ectasia (VAE) from clinics 1 and 2, and was divided into the frank ectasia eyes (VAE-E) and the fellow eyes with normal topography (VAE-NT). Normal topography was rigorously considered based on objective criteria[15] of Kmax less than 47.2 diopter, I-S (inferior-superior asymmetry[16]) less than 1.4, and KISA less than 60.[17] Table 2 summarizes the origin of patients included in the study.

Per the standards of the participating clinics, contact lens use was discontinued at least 3 weeks for rigid contact lens and 1 week for soft contact lens before the assessment. Patients with previous eye surgery, eye disease, extensive corneal scarring, or connective tissue disease, as well as pregnant or early puerperal women, were excluded.

• **PENTACAM TOMOGRAPHY:** All eyes were examined by rotating Scheimpflug corneal and anterior segment tomography (Pentacam HR; Oculus GmbH, Wetzlar, Germany). Image quality was checked, so that only cases with acceptable-quality images were included in the study. One experienced fellowship-trained corneal specialist (R.A.) reviewed all the cases so that they were correctly classified in the KC and VAE groups. The raw data (u12 files) was obtained from all cases so that the same customized software (version 1.20r118) was used to process all the export files from each center.

Pentacam keratometric values (central simulated Ks, KMax, and its vertical decentration, KMaxY); topometric indices (derived from the front surface: index of surface variance [ISV], index of vertical asymmetry [IVA], IS-value, keratoconus index [KI], central keratoconus index [CKI], index of height asymmetry [IHA], and index of height decentration [IHD]); and tomographic indices (anterior and posterior elevations at the thinnest point [AETP and PETP], anterior and posterior asphericity [AsphQFront and AsphQBack], pachymetry at the thinnest point [TP] and its dencentration [TPy], average and maximum and minimum pachymetric progression

TABLE 1. Surgical Procedure–Related Data of Patients who Had Laser-Assisted In Situ Keratomileusis Surgery

	Training Set								Test Set							
	Stable Train - Optical Express (UK) (N = 2980)				Post-LASIK Ectasia - International Pool (N = 71)				Stable Independent Test - Price Vision Group (USA) (N = 198)				Stable Independent Test - London Vision Clinic (UK) (N = 100)			
	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
Age	35.54	11.73	21	68	30.62	10.17	20	68	41.88	11.84	19	66	39.88	9.97	18	65
Flap thickness	131.81	18.25	120	160	133.94	19.16	110	160	115.71	5.54	110	160	101.60	13.43	80	130
Ablation depth	52.25	26.22	6	197	54.76	26.01	20	124	58.11	25.49	11.67	113.75	76.39	30.17	12	135
Corneal thickness	549.51	30.90	478	664	523.65	24.39	479	583	556.79	32.57	479	664	546.38	33.34	473	635
Flap technique	Femtossecond laser = 2099 (70.5%)		Microkeratome = 881 (29.5%)		Femtossecond laser = 26 (36.6%)		Microkeratome = 45 (63.4%)		Femtossecond laser = 198 (100%)				Femtossecond laser = 100 (100%)			

LASIK = laser-assisted in situ keratomileusis; Max = maximum; Min = minimum.

indices [PPI-Ave, PPI-Max, PPI-Min], average, maximum, minimum Ambrósio's relational thickness [ART-Ave, ART-Max, and ART-Min], and third version of Belin-Ambrósio display indices [BAD-D]) were computed in a spreadsheet.

• **ARTIFICIAL INTELLIGENCE ALGORITHMS AND VALIDATION METHODS:** Different machine learning techniques were used to analyze the tomographic data in order to identify the different patterns of each group. Models were built using regularized discriminant analysis (RDA), support vector machine (SVM), naïve Bayes (NB), neural networks (NN), and random forest (RF).

In brief, the RDA is a discriminant method that uses a regularized group covariance matrix. This avoids the stability problem of high correlated data in linear discriminant analysis. The SVM classifies the data by creating a high-dimensional hyperplane able to separate the elements of the different groups. The NB is based on Bayes' theorem; its main characteristic, whereby it is known as naïve, is that it considers the predictors to be independent, without any relation to each other. The NN models are based on the function of the neuron. They are formed by several individual units to which the sign is presented, processed and a binary response is reached. This process mimics the biological neuronal activation. Several of those units together form an artificial neural network. In the RF method, multiple decision trees are built and merged to improve accuracy of the prediction.[18]

To assess the generalizability and clinical validity of the models and their abilities to correctly classify new data, 2 steps of validation were used. The first was a holdout validation: the training set was randomly split into 2 data sets; the first comprised 70% of the total data set and was used to actually train the models and the other 30% was used to test the model accuracies. The second validation step was an independent test with cases that were not part of the training set.

• **STATISTICAL ANALYSIS:** The data distribution was assessed with the Kolmogorov-Smirnov goodness-of-fit test and almost all indices in the keratoconus group were non-normally distributed. Comparison among the groups was therefore performed with the nonparametric Kruskal-Wallis test, followed by the post hoc Dunn's test to compare each pair of groups. Receiver operating characteristic (ROC) curves were analyzed to determine the optimal cutoffs, sensitivity, and specificity. The area under the curve (AUC) was used as a measure of accuracy. DeLong's test was performed to pairwise compare different areas under the ROC curves (AUC), and the binomial exact was used for calculating the 95% confidence intervals (CI). It is important to note that when an analysis presents accuracy of 100%, it represents the studied population and an expected calculated extrapolation to other

TABLE 2. Origin of Patients Included in the Study					
	Clinic				
	Optical Express (UK) - Clinic 1	Instituto de Olhos Renato Ambrósio (Brazil) - Clinic 2	Vincieye Clinic (Italy) - Clinic 4	Price Vision Group (USA) - Clinic 4	London Vision Clinic (UK) - Clinic 5
KC	-	102	80	-	-
VAE-E	-	126	62	-	-
VAE-NT	-	126	62	-	-
Stable	2980	-	-	198	100

KC = keratoconus; VAE-E = frank ectatic eye in very asymmetric cases; VAE-NT = fellow normal-topography of very asymmetric ectasia cases.

populations in 95% of the cases will be on the range of the confidence interval.

Statistical analysis was accomplished with the MedCalc Statistical Software, version 17.7.2 (MedCalc Software bvba, Ostend, Belgium; http://www.medcalc.org; 2017), and all AI models were built using the R Core Team (2016), a language and environment for statistical computing (R Foundation for Statistical Computing, Vienna, Austria; https://www.R-project.org/).

RESULTS

THE MEAN AGE OF PATIENTS WAS 35.5 ± 11.7 YEARS IN STAble Train, 41.2 ± 11.2 years in Stable Test, 30.6 ± 10.1 years in PLE, 34.8 ± 11.5 years in KC, and 33.1 ± 13.4 years in the VAE group. Patient age differed significantly among the groups ($P < .0001$); in pairwise post hoc Dunn's test comparisons, only Stable Train vs KC and PLE vs VAE did not differ significantly ($P = .44$ and $P = .06$, respectively). The male-to-female ratios did not differ significantly among the groups ($P > .05$).

The comparison of the 6 groups showed statistically significant differences for all variables analyzed and for most of the pairwise comparisons ($P < .05$). The least significant differences were between the PLE and the VAE-NT groups. All parameters were statistically different between the Stable Train group and the PLE. K1, corneal astigmatism, and asphericity of the posterior surface were not statistically different between the Stable Test group and the PLE. Table 3 presents the descriptive statistics.

The index with the highest AUC from the tomographic examination was the final D score from the Belin-Ambósio display (BAD-D), 0.956 (sensitivity [se] = 89.3%, specificity [sp] = 90.7%) for the training set (PLE + KC), 0.845 (se = 64.8%, sp = 89.7%) for the detection of PLE alone, and 0.999 (95% CI 0.995–1.000, se = 98.9%, sp = 98.7%) for detection of KC alone. In the independent test set the optimized cutoff 1.505, used to detect PLE, classified correctly 55.3% of VAE-NT, 98.9% of VAE-E, and 95% of Stable Test. The PLE cases presented higher AUC in the anterior surface–related parameters, whereas the KC presented the best accuracy in the posterior surface–related parameters. Table 4 summarizes the ROC analysis and Figure 1 shows the separation plots of BAD-D.

Five different AI models were trained. The accuracies were measured with the AUC. The RF had the highest accuracy, 0.992. Compared with the RF model, the AUC was significantly lower in the other 4 models: the AUC of Naïve Bayes was 0.961 ($P = .01$), NN was 0.959 ($P = .01$), SVM was 0.94 ($P = .003$), and RDA was 0.886 ($P < .0001$). Figure 2 shows the ROC curves of the internal cross-validation of some of the AI models and the BAD-D.

The Pentacam Random Forest Index (PRFI) was then defined. Compared with BAD-D, the PRFI had a significantly higher AUC of 0.992 in the training set (se = 94.2%, sp = 98.8%, $P = .006$). For the detection of PLE the AUC was statistically higher with the PRFI, 0.966 (se = 80%, sp = 96.8%, $P = .002$), whereas for detection of KC no difference was found for this sample (AUC = 0.999, 95% CI 0.996–1.000, se = 100%, sp = 99.7%; $P = .07$). A considerable improvement was seen in the VAE detection; the optimized cutoff of 0.125 used to detect PLE correctly classified 85.2% of VAE-NT and all VAE-E cases. Regarding the Stable Test, 96.6% were correctly classified. Figure 3 shows the separation plots of PRFI.

DISCUSSION

THE PREOPERATIVE SCREENING PROCESS FOR REFRACTIVE surgery candidates is important to rule out predisposing conditions to iatrogenic ectasia.[4] The most important independent risk factor described is corneal preexisting alteration that resembles ectasia (KC or pellucid marginal corneal degeneration).[1,19,20] Even though the clinically evident cases are easy to detect by retinoscopy, slit-lamp examination, and topographic findings, the early stages of ectasia are hard to detect, because of the high similarity to normal corneas. Another issue is the lack of consensus in defining these cases, so that often the results of different studies cannot be directly compared. Several terms have been employed to describe corneas with early forms of

TABLE 3. Descriptive Statistics of Train and Test Sets

	Stable Train Set (N = 2980)				Stable Independent Test Set (N = 298)				Post-LASIK Ectasia Train Set (N = 71)			
	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
Age[a]	35.54	11.73	0.21	68.70	41.21	11.27	18.00	66.40	30.62	10.17	19.98	68.43
K1[b]	43.12	1.41	38.3	48.2	43.26	1.52	39.4	47.3	43.51	1.41	40.3	46.3
K2	44.15	1.46	39.6	48.9	44.36	1.53	40.7	48.3	44.77	1.42	41.1	48.3
Astig[a,b]	1.03	0.70	0	4.8	1.10	0.77	0	5	1.26	0.83	0.1	3.8
K Max Front	44.79	1.52	40.5	50.6	45.01	1.59	41.3	49.9	45.67	1.59	41.6	49.7
ISV[a]	16.00	5.42	5	82	16.16	5.72	7	45	21.54	6.81	10	40
IHD[a]	0.01	0.00	0	0.043	0.01	0.00	0.001	0.017	0.01	0.01	0.001	0.048
KISA	14.72	18.28	0.333	245.243	15.20	18.39	0.333	163.054	30.53	37.42	0.432	148.588
I-S Value[a]	0.08	0.62	−3.279	3.673	0.29	0.57	−1.501	1.834	0.78	0.74	−1.249	3.219
Ele F BFS 8 mm Thinnest	1.67	1.74	−13	16	1.36	1.47	−3	5	4.21	2.29	−1	13
Asph. Q Front (30 deg)[a]	−0.22	0.11	−0.77	0.13	−0.21	0.12	−0.55	0.12	−0.32	0.16	−0.72	0.12
Ele B BFS 8 mm Thinnest	4.42	5.18	−8	74	3.96	3.64	−3	15	8.94	7.52	−3	34
Asph. Q Back (30 deg)[b]	0.01	0.27	−1.63	0.8	−0.04	0.20	−0.77	0.48	−0.10	0.34	−1.1	0.72
Pachy Apex[a]	547.08	30.67	473	665	553.61	33.15	473	664	528.39	24.04	484	584
Pachy Min[a]	544.13	30.69	469	663	551.54	33.41	468	663	523.65	24.39	479	583
ART Avg[a]	610.10	116.01	0	1405	644.62	112.66	385	1257	491.55	85.02	264	682
ART Max[a]	489.00	95.97	0	966	522.50	98.65	273	878	388.24	78.93	162	574
BAD D[a]	0.65	0.70	−1.6	4.93	0.47	0.62	−1.59	2.01	1.82	0.97	−0.11	5.13
PRFI[a]	0.02	0.05	0	0.868	0.02	0.04	0	0.332	0.46	0.29	0	0.92
	KC Train Set (N = 182)				VAE-NT Independent Test Set (N = 188)				VAE-E Independent Test Set (N = 188)			
	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
Age[a]	34.83	11.57	12.70	72.57	33.09	13.44	13.17	83.21	33.09	13.44	13.17	83.21
K1[b]	46.29	4.83	35.4	75.8	42.81	1.28	39.6	45.7	44.39	4.03	33.4	67.1
K2	49.36	5.45	42.3	77.1	43.85	1.34	40.4	46.5	47.98	4.47	41	69.4
Astig[a,b]	3.06	1.87	0.1	10.1	1.04	0.69	0.1	4.2	3.59	2.22	0.3	13.6
K Max Front	54.83	8.01	45.2	101.2	44.63	1.37	41.3	47.1	52.82	6.59	44.5	82.9
ISV[a]	81.30	43.83	20	338	19.22	5.83	8	42	72.36	36.85	24	223
IHD[a]	0.12	0.08	0.016	0.559	0.01	0.01	0.001	0.029	0.08	0.06	0.012	0.36
KISA	2866.84	13584.34	2.297	173021.73	13.66	13.41	0.333	54.138	1848.78	9796.56	1.126	124404.039
I-S Value[a]	6.00	4.32	−2.6	33.69	0.54	0.55	−1.004	1.384	4.53	3.26	−2.944	18.546
Ele F BFS 8 mm Thinnest	20.15	14.11	−50	72	3.18	2.01	−2	9	18.99	12.29	−8	83
Asph. Q Front (30 deg)[a]	−0.69	0.43	−2.55	0.52	−0.33	0.15	−0.76	0.06	−0.64	0.44	−2.21	0.56
Ele B BFS 8 mm Thinnest	58.03	132.56	2	1805	10.14	7.29	−5	50	45.04	27.67	−6	245
Asph. Q Back (30 deg)[b]	−55.66	741.12	−9999	1.57	−0.21	0.29	−1.14	1.38	−0.55	0.63	−2.42	1.3
Pachy Apex[a]	486.09	128.83	209	2113	521.21	32.76	435	608	490.66	41.11	359	580
Pachy Min[a]	463.20	45.15	173	553	513.68	33.73	403	602	475.11	42.68	341	567
ART Avg[a]	252.09	94.58	0	653	486.73	101.42	115	713	291.27	122.01	0	871
ART Max[a]	171.09	69.03	0	440	368.27	88.39	79	612	200.54	91.65	0	653
BAD D[a]	9.16	13.03	1.46	168.12	1.78	0.99	−0.22	7.68	7.09	4.29	0.91	29.22
PRFI[a]	0.98	0.05	0.546	1	0.43	0.27	0	0.942	0.91	0.09	0.604	1

ART Avg = average Ambrósio relational thickness; ART Max = maximum Ambrósio relational thickness; Asph. Q Back (30 deg) = back surface asphericity at 30 degrees; Asph. Q Front (30 deg) = front surface asphericity at 30 degrees; Astig = corneal astigmatism in 3-mm zone; BAD-D = total deviation of Belin-Ambrósio display; Ele B BFS 8 mm Thinnest = back surface elevation at the thinnest point using the 8-mm best fit sphere; Ele F BFS 8 mm Thinnest = front surface elevation at the thinnest point using the 8-mm best fit sphere; IHD = index of height decentration; ISV = index of surface variance; I-S Value = inferior-superior asymmetry; K1 = flat meridian curvature; K2 = steep meridian curvature; KC = keratoconus; KISA = index with the keratometry (K) value, inferior-superior (I-S) value, relative skewing of the steepest radial axes (SRAX), and keratometric astigmatism (AST); K Max Front = maximum curvature of front surface; Pachy Apex = corneal thickness at the apex; Pachy Min = minimum corneal thickness; PRFI = Pentacam Random Forest Index; SD = standard deviation; VAE-E = ectasia eye of very asymmetric cases; VAE-NT = normal topography eye of very asymmetric cases.

[a]Nonsignificant different between post-LASIK ectasia and VAE-NT.

[b]Nonsignificant different between post-LASIK ectasia and Stable Test.

TABLE 4. Receiver Operating Characteristic Curve Analysis

	Train Data Set											
	Stable Versus All Ectasia Cases				Stable Versus KC				Stable Versus PLE			
	AUC	Sensitivity	Specificity	Cutoff	AUC	Sensitivity	Specificity	Cutoff	AUC	Sensitivity	Specificity	Cutoff
K1	0.714	0.494	0.860	44.65	0.767	0.522	0.919	45.15	0.579	0.310	0.843	44.55
K2	0.808	0.597	0.888	45.95	0.877	0.709	0.916	46.25	0.631	0.704	0.543	44.25
Astig	0.783	0.621	0.849	1.65	0.861	0.747	0.849	1.65	0.581	0.282	0.875	1.75
K Max Front	0.888	0.684	0.942	47.25	0.977	0.868	0.966	47.65	0.660	0.761	0.532	44.85
ISV	0.925	0.798	0.944	25.5	0.995	0.945	0.987	32.5	0.745	0.577	0.798	19.5
IHD	0.915	0.822	0.941	0.0135	1.000	0.995	0.988	0.0205	0.698	0.493	0.819	0.0095
KISA	0.869	0.704	0.986	58.89	0.967	0.890	0.987	60.65	0.617	0.225	0.986	58.89
I-S Value	0.923	0.783	0.971	1.13	0.983	0.967	0.991	1.49	0.770	0.789	0.640	0.33
Ele F BFS 8 mm Thinnest	0.930	0.810	0.965	4.5	0.974	0.951	0.965	4.5	0.820	0.775	0.711	2.5
Asph. Q Front (30 deg)	0.836	0.660	0.928	−0.385	0.887	0.780	0.941	−0.395	0.704	0.662	0.706	−0.285
Ele B BFS 8 mm Thinnest	0.906	0.771	0.934	12.5	0.988	0.945	0.979	16.5	0.696	0.634	0.648	5.5
Asph. Q Back (30 deg)	0.802	0.593	0.897	−0.335	0.876	0.769	0.873	−0.295	0.610	0.535	0.706	−0.105
Pachy Apex	0.852	0.664	0.877	511.5	0.920	0.780	0.895	509.5	0.676	0.648	0.618	535.5
Pachy Min	0.882	0.735	0.881	508.5	0.955	0.863	0.916	503.5	0.694	0.648	0.656	529.5
ART Avg	0.939	0.822	0.949	455.5	0.992	0.984	0.968	435.5	0.803	0.732	0.780	523.5
ART Max	0.939	0.810	0.956	349.5	0.995	0.978	0.983	317.5	0.797	0.761	0.719	433.5
BAD D	0.956	0.893	0.907	1.545	0.999	0.989	0.987	2.33	0.845	0.648	0.897	1.505
PRFI[a]	0.992	0.942	0.988	0.216	0.999	1.000	0.997	0.478	0.966	0.800	0.968	0.125

	Test Data Set (Optimized Cutoff to Detect PLE)				
	AUC		Sensitivity		Specificity
	VAE-NT Versus Stable	VAE-E Versus Stable	VAE-NT	VAE-E	Stable
K1	0.580	0.575	0.920	0.335	0.211
K2	0.588	0.813	0.622	0.878	0.503
Astig	0.488	0.888	0.106	0.809	0.829
K Max Front	0.561	0.957	0.564	0.984	0.500
ISV	0.665	0.997	0.404	1.000	0.765
IHD	0.779	0.999	0.532	1.000	0.862
KISA	0.504	0.927	0.000	0.787	0.983
I-S-Value	0.635	0.949	0.707	0.957	0.507
Ele F BFS 8 mm Thinnest	0.764	0.965	0.644	0.963	0.792
Asph. Q Front (30 deg)	0.745	0.848	0.633	0.798	0.765
Ele B BFS 8 mm Thinnest	0.803	0.985	0.777	0.984	0.701
Asph. Q Back (30 deg)	0.719	0.779	0.702	0.750	0.671
Pachy Apex	0.762	0.893	0.691	0.872	0.698
Pachy Min	0.793	0.932	0.707	0.899	0.745
ART Avg	0.862	0.983	0.654	0.957	0.879
ART Max	0.890	0.989	0.777	0.984	0.839
BAD-D	0.893	0.997	0.553	0.989	0.950
PRFI[a]	0.968	1	0.852	1	0.966

ART Avg = average Ambrósio relational thickness; ART Max = maximum Ambrósio relational thickness; Asph. Q Back (30 deg) = back surface asphericity at 30 degrees; Asph. Q Front (30 deg) = front surface asphericity at 30 degrees; Astig = corneal astigmatism in 3-mm zone; BAD-D = total deviation of Belin-Ambrósio display; Ele B BFS 8 mm Thinnest = back surface elevation at the thinnest point using the 8-mm best fit sphere; Ele F BFS 8 mm Thinnest = front surface elevation at the thinnest point using the 8-mm best fit sphere; IHD = index of height decentration; ISV = index of surface variance; I-S Value = inferior-superior asymmetry; K1 = flat meridian curvature; K2 = steep meridian curvature; KC = keratoconus; KISA = index with the keratometry (K) value, inferior-superior (I-S) value, relative skewing of the steepest radial axes (SRAX), and keratometric astigmatism (AST); K Max Front = maximum curvature of front surface; Pachy Apex = corneal thickness at the apex; Pachy Min = minimum corneal thickness; PLE = post-LASIK ectasia; PRFI = Pentacam Random Forest Index; SD = standard deviation; VAE-E = ectasia eye of very asymmetric cases; VAE-NT = normal topography eye of very asymmetric cases.

[a]Cross-validation results for Training Set.

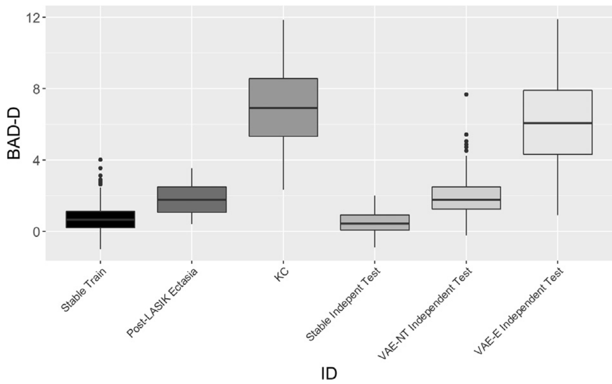

FIGURE 1. Comparison of total deviation of Belin-Ambrósio display values (BAD-D) distribution among all groups of stable and ectatic eyes. BAD-D = total deviation of Belin-Ambrósio display; KC = keratoconus; VAE-E = ectasia eye of very asymmetric cases; VAE-NT = normal topography eye of very asymmetric cases.

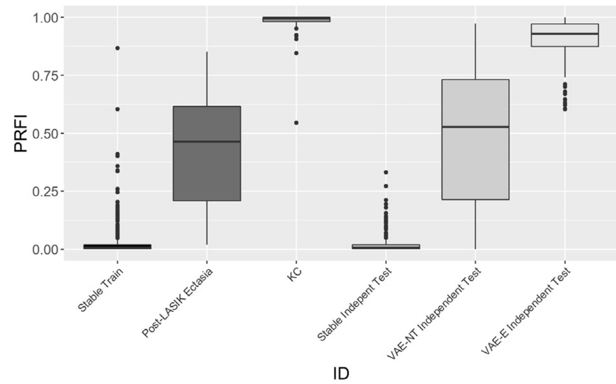

FIGURE 3. Comparison of distribution of Pentacam Random Forest Index (PRFI) values among all groups of stable and ectatic eyes. KC = keratoconus; PRFI = Pentacam Random Forest Index; VAE-E = ectasia eye of very asymmetric cases; VAE-NT = normal topography eye of very asymmetric cases.

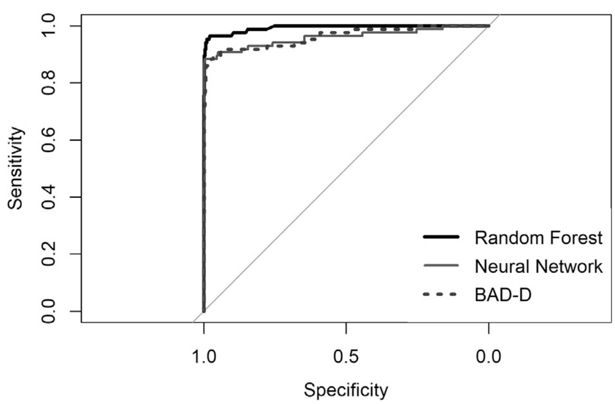

FIGURE 2. Comparison of artificial intelligence models and total deviation of Belin-Ambrósio display value (BAD-D) with receiver operating characteristic (ROC) curve analysis.

the disease, including forme fruste keratoconus (FFKC), keratoconus suspect (KCS), and preclinical and subclinical keratoconus. Amsler was the first to describe the "forme fruste" cases.[21] His definition was based on Placido's disk image reflection only and included patterns that after the development of computerized corneal videokeratography were considered diagnostic of the manifest disease. Klyce used the term "forme fruste" for topographically normal fellow eyes of patients with clinically evident unilateral keratoconus and suggested that it should be differentiated from KCS cases with characteristic topographic signs.[22] To avoid the overlapping definitions of FFKC and KCS present in many studies, we adopted the term "VAE-NT" to address the normal topography eye in very asymmetrical cases.

Whereas the suspect cases can be detected because they already show topographic signs of the disease, the most pressing clinical challenge is detection of susceptible cases, with no disease measurable by current screening methods but with weak corneas that can progress to clinically evident disease.[23] The best study design is to compare the preoperative data of patients that remained stable after LASIK surgery with data from patients that developed iatrogenic ectasia. So we included both groups in the training population to develop an AI model better able to separate them. The KC group was also included in the training set to evaluate the coherence of the findings considering keratoconus as a progressive disease.[24,25] Because the preoperative tomography data of the PLE is hard to find, we decided to include all the cases in the training set, so we could assess all its variability with the AI modeling. The VAE-NT was used to test the model accuracy, because this is the group with the highest similarity with the PLE. However, it is not a perfect surrogate. One of its limitations is the possibility that cases of true unilateral ectasia exist.[26] Another limitation is the slight tendency that the ectasia cases present more alterations in the anterior surface than the VAE-NT and the KC. Bühren and associates assessed the preoperative data of 10 eyes that developed PLE; the authors also found that the alterations in the ectasia group were different from the pattern found in subclinical keratoconus.[27] However, though the values of posterior elevation at the thinnest point were lower in the PLE than in the VAE-NT, they were still higher than in the stable corneas.

In our study we trained a machine-learning algorithm of random forest, the PRFI. Data from 3 continents were used to consider ethnic differences and this resulted in a large study cohort. To avoid overfitting and assess the external validity, a holdout validation method was performed along with independent tests. The accuracy of the new index was an improvement over previous tomographic indices. The best tomographic index previously available was the

BAD-D, which correctly classified only 55.3% of the PLE, whereas the PRFI correctly classified 80%. Despite the improved the accuracy, the misclassification of 20% of the cases suggests the need for further research to develop a better understanding of corneal structure, including data from the epithelial thickness,[28,29] which may be distinguished as segmental tomography.[30] The biomechanical status of the cornea also is very promising to add significant value for such analysis.[31] For example, the Corneal Biomechanical Index[32] provides the ability to detect ectasia in eyes with relatively normal corneal tomography.[33] Nevertheless, the integration of corneal tomography and biomechanical data, as was done with the Tomographic-Biomechanical Index, has been demonstrated to improve the ability to detect these very early cases.[34]

In the detection of the VAE-NT the PRFI presented sensitivity of 85.2% and specificity of 96.6% considering only the independent test set. Some authors have applied different AI methods in order to detect the VAE-NT. Saad and Gatinel described a linear discriminant–based model that was developed with another corneal tomographer (Orbscan II; Technolas, Munich, Germany), with higher sensitivity (93%) and lower specificity (92%).[35] It was further validated in 2 independent populations. In one study published by the authors, the sensitivity and specificity were similar to the original sample, 92% and 96%, respectively.[36] In the other, a different ethnic group was evaluated and a decrease in sensitivity to 70.8% was observed, with the maintenance of specificity (98.1%).[37] The difference of the sensitivity in both studies may be explained in part by overfitting of the studied population that can overestimate results. In our data set, the VAE-NT were all independently tested, so they were not used in the training set. This dramatically reduces the risk of overfitting and thus gives a more realistic result and estimate of sensitivity. Another strength of our study was the large number of cases. Previously, a total of 454 cases were analyzed with 88 VAE-NT, whereas our study analyzed 3693 cases with 188 VAE-NT. The larger sample size and broader representation of different populations also helps to avoid overfitting and assure more reliable results.

The study by Smadja and associates used single DT methods to detect ectatic corneas based on another tomographer, the Galilei Dual Scheimpflug Analyzer (Ziemer Ophthalmic Systems AG, Port, Switzerland). The separation between normal and KC obtained sensitivity and specificity higher than 99%. The detection of VAE-NT achieved sensitivity of 93.6% and specificity of 97.2% that dropped to 90% and 86%, respectively, after pruning.[38] However, the authors used 2 separate models to detect KC and VAE-NT. In our opinion, including the KC control group is important to assess the coherence and avoid overfitting of the VAE-NT model. We believe that because of this, Smadja associates' study is not directly comparable with ours.

Two studies used different AI approaches to identify VAE-NT cases using the Pentacam HR. Ruiz Hidalgo and associates reported an SVM model that achieved a slightly lower AUC of 0.922 with sensitivity of 79.1% and specificity of 97.9% in the detection of the VAE-NT.[39] Kovács and associates approached the VAE-NT detection problem using neural networks and bilateral data of patients. They had an AUC of 0.96 with sensitivity of 92% and specificity of 85%,[40] which was similar to what we obtained with PRFI (AUC = 0.968). However no independent test set was used, which was identified by the authors as a study limitation.

Arbelaez and associates used the SVM model to detect subclinical keratoconus and achieved sensitivity of 92% with specificity of 97.7%.[41] However, their results cannot be directly compared with ours because their definition of subclinical keratoconus included first signs of the disease and thus does not match our VAE-NT definition. Bae and associates, with the single indices of Pentacam HR, reached the best accuracy in detection of the VAE-NT with the asymmetry index ST-IN (superotemporal–inferonasal asymmetry), AUC = 0.809.[42] However, this is lower than the AUC we obtained in this study with the PRFI (0.968). We need to keep in mind that ectasia susceptibility depends not only on corneal fragility (resemblance with VAE-NT) but also with surgical parameters (ie, flap/ablation depth), which still need to be better understood.[5] Thereby, along with the improvement in accuracy for detecting ectatic or susceptible cases, the integration with the data that resembles the biomechanical impact from the procedure is fundamental to individually assess ectasia risk prior to laser vision correction.[43]

In conclusion, we have described a machine learning parameter model that uses tomographic data to detect ectasia and its susceptibility. The PRFI achieved excellent accuracy in detection of KC and a superior accuracy in the detection of the PLE and VAE-NT in independent tests than the parameters previously reported in the literature. However, the accuracy in the early and susceptible keratoconus is still subject for further improvements in the algorithm with the integration of other shape, biomechanical, and surgical parameters.

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REFERENCES

1. Seiler T, Koufala K, Richter G. Iatrogenic keratectasia after laser in situ keratomileusis. *J Refract Surg* 1998;14(3):312–317.
2. Seiler T, Quurke AW. Iatrogenic keratectasia after LASIK in a case of forme fruste keratoconus. *J Cataract Refract Surg* 1998;24(7):1007–1009.
3. Binder PS, Lindstrom RL, Stulting RD, et al. Keratoconus and corneal ectasia after LASIK. *J Refract Surg* 2005;21(6):749–752.
4. Ambrosio R Jr, Randleman JB. Screening for ectasia risk: what are we screening for and how should we screen for it? *J Refract Surg* 2013;29(4):230–232.
5. Santhiago MR, Smadja D, Gomes BF, et al. Association between the percent tissue altered and post-laser in situ keratomileusis ectasia in eyes with normal preoperative topography. *Am J Ophthalmol* 2014;158(6):87–95 e1.
6. Ambrosio R Jr, Dawson DG, Belin MW. Association between the percent tissue altered and post-laser in situ keratomileusis ectasia in eyes with normal preoperative topography. *Am J Ophthalmol* 2014;158(6):1358–1359.
7. Comaish IF, Lawless MA. Progressive post-LASIK keratectasia: biomechanical instability or chronic disease process? *J Cataract Refract Surg* 2002;28(12):2206–2213.
8. Kim H, Joo CK. Measure of keratoconus progression using Orbscan II. *J Refract Surg* 2008;24(6):600–605.
9. Ambrosio R Jr, Klyce SD, Wilson SE. Corneal topographic and pachymetric screening of keratorefractive patients. *J Refract Surg* 2003;19(1):24–29.
10. Klein SR, Epstein RJ, Randleman JB, Stulting RD. Corneal ectasia after laser in situ keratomileusis in patients without apparent preoperative risk factors. *Cornea* 2006;25(4):388–403.
11. Ambrosio R Jr, Dawson DG, Salomao M, Guerra FP, Caiado AL, Belin MW. Corneal ectasia after LASIK despite low preoperative risk: tomographic and biomechanical findings in the unoperated, stable, fellow eye. *J Refract Surg* 2010;26(11):906–911.
12. Ambrósio R Jr, Belin MW. Enhanced screening for ectasia risk prior to laser vision correction. *Int J Kerat Ect Cor Dis* 2017;6(1):22–33.
13. Maeda N, Klyce SD, Smolek MK, Thompson HW. Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci* 1994;35(6):2749–2757.
14. Maeda N, Klyce SD, Smolek MK. Neural network classification of corneal topography. Preliminary demonstration. *Invest Ophthalmol Vis Sci* 1995;36(7):1327–1335.
15. Ramos IC, Correa R, Guerra FP, et al. Variability of subjective classifications of corneal topography maps from LASIK candidates. *J Refract Surg* 2013;29(11):770–775.
16. Rabinowitz YS, McDonnell PJ. Computer-assisted corneal topography in keratoconus. *Refract Corneal Surg* 1989;5(6):400–408.
17. Rabinowitz YS, Rasheed K. KISA% index: a quantitative videokeratography algorithm embodying minimal topographic criteria for diagnosing keratoconus. *J Cataract Refract Surg* 1999;25(10):1327–1335.
18. Amancio DR, Comin CH, Casanova D, et al. A systematic comparison of supervised classifiers. *PLoS One* 2014;9(4):e94137.
19. Randleman JB, Trattler WB, Stulting RD. Validation of the Ectasia Risk Score System for preoperative laser in situ keratomileusis screening. *Am J Ophthalmol* 2008;145(5):813–818.
20. Binder PS, Lindstrom RL, Stulting RD, et al. Keratoconus and corneal ectasia after LASIK. *J Cataract Refract Surg* 2005;31(11):2035–2038.
21. Amsler M. [The "forme fruste" of keratoconus]. *Wiener klinische Wochenschrift* 1961;73:842–843.
22. Klyce SD. Chasing the suspect: keratoconus. *Br J Ophthalmol* 2009;93(7):845–847.
23. Li X, Rabinowitz YS, Rasheed K, Yang H. Longitudinal study of the normal eyes in unilateral keratoconus patients. *Ophthalmology* 2004;111(3):440–446.
24. Krachmer JH, Feder RS, Belin MW. Keratoconus and related noninflammatory corneal thinning disorders. *Surv Ophthalmol* 1984;28(4):293–322.
25. Rabinowitz YS. Keratoconus. *Surv Ophthalmol* 1998;42(4):297–319.
26. Imbornoni LM, Padmanabhan P, Belin MW, Deepa M. Long-term tomographic evaluation of unilateral keratoconus. *Cornea* 2017;36(11):1316–1324.
27. Buhren J, Schaffeler T, Kohnen T. Preoperative topographic characteristics of eyes that developed postoperative LASIK keratectasia. *J Refract Surg* 2013;29(8):540–549.
28. Silverman RH, Urs R, Roychoudhury A, Archer TJ, Gobbe M, Reinstein DZ. Epithelial remodeling as basis for machine-based identification of keratoconus. *Invest Ophthalmol Vis Sci* 2014;55(3):1580–1587.
29. Li Y, Chamberlain W, Tan O, Brass R, Weiss JL, Huang D. Subclinical keratoconus detection by pattern analysis of

corneal and epithelial thickness maps with optical coherence tomography. *J Cataract Refract Surg* 2016;42(2):284–295.
30. Salomao MQ, Hofling-Lima AL, Lopes BT, et al. Role of the corneal epithelium measurements in keratorefractive surgery. *Curr Opin Ophthalmol* 2017;28(4):326–336.
31. Roberts CJ, Dupps WJ Jr. Biomechanics of corneal ectasia and biomechanical treatments. *J Cataract Refract Surg* 2014;40(6): 991–998.
32. Vinciguerra R, Ambrósio R Jr, Elsheikh A, et al. Detection of keratoconus with a new corvis ST biomechanical index. *J Refract Surg* 2016;32(12):803–810.
33. Vinciguerra R, Ambrosio R Jr, Roberts CJ, Azzolini C, Vinciguerra P. Biomechanical characterization of subclinical keratoconus without topographic or tomographic abnormalities. *J Refract Surg* 2017;33(6):399–407.
34. Ambrosio R Jr, Lopes BT, Faria-Correia F, et al. Integration of Scheimpflug-based corneal tomography and biomechanical assessments for enhancing ectasia detection. *J Refract Surg* 2017;33(7):434–443.
35. Saad A, Gatinel D. Topographic and tomographic properties of forme fruste keratoconus corneas. *Invest Ophthalmol Vis Sci* 2010;51(11):5546–5555.
36. Saad A, Gatinel D. Validation of a new scoring system for the detection of early forme of keratoconus. *Int J Kerat Ect Cor Dis* 2012;1(2):100–108.
37. Chan C, Ang M, Saad A, et al. Validation of an objective scoring system for forme fruste keratoconus detection and post-LASIK ectasia risk assessment in Asian eyes. *Cornea* 2015;34(9):996–1004.
38. Smadja D, Touboul D, Cohen A, et al. Detection of subclinical keratoconus using an automated decision tree classification. *Am J Ophthalmol* 2013;156(2):237–246 e1.
39. Ruiz Hidalgo I, Rodriguez P, Rozema JJ, et al. Evaluation of a machine-learning classifier for keratoconus detection based on Scheimpflug tomography. *Cornea* 2016;35(6): 827–832.
40. Kovacs I, Mihaltz K, Kranitz K, et al. Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. *J Cataract Refract Surg* 2016;42(2): 275–283.
41. Arbelaez MC, Versaci F, Vestri G, Barboni P, Savini G. Use of a support vector machine for keratoconus and subclinical keratoconus detection by topographic and tomographic data. *Ophthalmology* 2012;119(11):2231–2238.
42. Bae GH, Kim JR, Kim CH, Lim DH, Chung ES, Chung TY. Corneal topographic and tomographic analysis of fellow eyes in unilateral keratoconus patients using Pentacam. *Am J Ophthalmol* 2014;157(1):103–109 e1.
43. Ambrosio R Jr, Ramos I, Lopes B, et al. Assessing ectasia susceptibility prior to LASIK: the role of age and residual stromal bed (RSB) in conjunction to Belin-Ambrósio deviation index (BAD-D). *Rev Bras Oftalmol* 2014;73(2): 75–80.